The Exact Computability of Two Adversarial Prediction Measures

28 October 2026
Working manuscript

Abstract

We determine the oracle complexity of Hibbard's original intelligence measure in the representation used by Alexander and Hibbard: total binary predictors are given by ordinary Turing-machine indices, and evaders are grouped by eventual strict worst-case runtime domination. The extended-natural-valued measure and its strict comparator have exact Turing degree $0'''$. Every sufficiently high fixed threshold, and the predicate of infinite measure, is Π^0_3-complete on the total-predictor promise. We then replace the primitive-recursive list by the canonical repetition-free list of all total computable functions, ordered by least program index. Every resulting measure value is finite, but the measure still has exact degree $0'''$. Thus this specified all-total-function variant does not require a stronger oracle than the original measure. A robust prediction lemma handles the feedback created by modified predictions; a uniform threshold formula prevents an unnecessary extra jump when the list itself is noncomputable.

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1 Intelligence measures as represented mathematical functions

In adversarial sequence prediction, a predictor tries to match the next bit chosen by an evader. Each side sees the other side's previous outputs. Hibbard's measure assigns a predictor a level

according to the runtime classes of evaders it eventually learns. Alexander and Hibbard [1] separate this underlying idea from the particular enumeration used to organize growth rates. After proving noncomputability, they ask for more precise degrees and conjecture that replacing primitive recursive functions by all total computable functions would require a strictly stronger oracle.

These questions depend on representation. A predictor is a total function, but an input to an algorithm is a program index. A list of all total computable functions cannot have a computable total evaluator, although each function in it is individually computable. The measure can also take the extra value ∞. We make each of these choices explicit.

Two main results follow. The original measure requires exactly the third Turing jump. A specified canonical all-total-function variant requires the same jump, even though its enumeration is noncomputable and it has no infinite values. The latter answers a precise instance of the proposed strict increase negatively; it does not classify every possible ordering of the total computable functions.

2 The game, the machines and the measures

Fix an acceptable numbering of ordinary Turing programs on finite binary strings, with an effective bounded simulation. A total binary program computes a map $\{0,1\}^{<\omega} \to \{0,1\}$. We use the standard input-tape convention in which a program can inspect its first k input cells and stop without scanning the remaining input. There is no mandatory input-reading cost proportional to its length.

Given a total predictor p and total evader e, define their unique play by

$$y_n = p(x_0 \dots x_{n-1}), \qquad x_n = e(y_0 \dots y_{n-1}).$$

Both empty histories occur at round zero. The evader sees the *predictions*, not its own previous evasions. The predictor *learns* e if $x_n = y_n$ for every sufficiently large n. For a total evader, let $t_e(n)$ be its maximum running time on all 2^n binary inputs of length n.

For an ordinary total computable bound $f : \mathbb{N} \to \mathbb{N}$, put

$$E_f = \{e : e \text{ is total binary and } t_e(n) < f(n) \text{ for all sufficiently large } n\},$$

and write $T(p, f)$ for the assertion that p learns every $e \in E_f$. The strict inequality is part of the contract. A finite initial segment of exceptions is allowed.

Let $g_1, g_2, \dots$ be a listing of growth functions and define

$$f_m(n) = \max_{1 \le i \le m,\ 0 \le j \le n} g_i(j), \qquad m \ge 1. \tag{2.1}$$

The bounds are pointwise nondecreasing in m. Define

$$H_g(p) = \begin{cases} 0, & \text{if } T(p, f_1) \text{ fails,} \\ \max\{m : T(p, f_m)\}, & \text{if this set is nonempty and bounded,} \\ \infty, & \text{if every } T(p, f_m) \text{ holds.} \end{cases}$$

The initial-segment property makes these cases exhaustive.

For the original measure, use the fixed effective primitive-recursive listing of [1, Definition 7]. What is needed here is an effective total evaluator for that listing, its coverage of all primitive recursive functions, and hence coverage of $n \mapsto n+1$. Call its measure H_{PR}.

All oracle computations of a measure below have the promise that their input index is a total binary predictor. They must halt and return the right answer on that domain. A machine computing an extended-natural-valued measure outputs a natural number or a distinct symbol for ∞. Statements of exact degree use this representation.

2.1 The complexity statements

Write $0', 0'', 0'''$ for successive Turing jumps. We use the standard arithmetical-hierarchy fact that a recursive-matrix Π^0_3 predicate is decidable in $0'''$, and that complete such predicates compute $0'''$.

Theorem 2.1. *There is a fixed coding-dependent m_0 such that:*

1. *for every fixed $m \geq m_0$, the predicate $H_{\mathrm{PR}}(p) \geq m$ is Π^0_3-complete on total predictor indices;*
2. *$H_{\mathrm{PR}}(p) = \infty$ is Π^0_3-complete on the same promise;*
3. *H_{PR} and the strict comparison relation between two such values have exact oracle degree $0'''$.*

The reductions witnessing hardness uniformly produce total predictors.

Theorem 2.2. *List all total computable functions without repetitions in order of their least program indices, and use* (2.1)*. For this canonical list, every $H_g(p)$ is finite and H_g has exact oracle degree $0'''$ on total predictor indices. Every sufficiently high fixed threshold is Π^0_3-complete on that promise.*

The hardness statements for fixed thresholds in the second theorem are nonuniform in the list position: a fixed position has a fixed ordinary bound program which can be hardcoded. The theorem does not assert that these bound indices can be found computably from their positions.

3 A uniform Π^0_3 threshold formula

Lemma 3.1. *Uniformly in an ordinary program index for a total computable f, membership $e \in E_f$ has a recursive-matrix Σ^0_2 definition.*

Proof. An existential witness consists of a cutoff N and a common finite halting certificate for all inputs of length at most N. Check that these computations halt with binary outputs. For each longer input x and each stage s, require: if the f program on $|x|$ has halted by s with value v, then the e program on x halts with a binary output in fewer than v steps. This is a bounded decidable test. Since f is total, the guard eventually activates at each input length. The universal tests therefore impose exactly eventual strict runtime domination and total binary output beyond the cutoff; the finite certificate covers all remaining inputs. Thus

$$e \in E_f \quad \Longleftrightarrow \quad \exists a \, \forall b \; R(e, f, a, b)$$

for a recursive predicate R, on the total-f promise. □

Lemma 3.2. *There is a recursive predicate S such that, for total binary p, e, p learns e exactly when $\exists c \forall d\ S(p, e, c, d)$.*

Proof. Let c be an error cutoff. The number d may encode a finite play, halting times for all its computations, and a designated error round at least c. Reject exactly when this is a valid computation certificate for such an error. Validity is decidable by bounded simulation. Every actual error has a finite certificate, and every valid certificate describes an actual initial play by determinism. Totality ensures the actual play is infinite. Absence of all post-cutoff error certificates is therefore eventual learning. □

Proposition 3.3. *Uniformly in total binary p and an ordinary total bound program f, $T(p, f)$ has a recursive-matrix Π^0_3 definition.*

Proof. Using the two preceding lemmas, write

$$\forall e, a\ \exists b, c\ \forall d\ \big(\neg R(e, f, a, b)\ \vee\ S(p, e, c, d)\big). \tag{3.1}$$

If a does not certify membership in E_f, one counterexample b makes the disjunction true. If it does, a learning witness c is required. Partial or nonbinary evaders cannot satisfy the membership antecedent, so their behavior in the learning formula is irrelevant. This proves equivalence and the claimed uniformity. □

The uniformity matters later. Computing an ordinary index for f with an oracle and then applying (3.1) does not require three jumps *relative* to that oracle. The formula already has a recursive matrix with the ordinary index as a parameter.

For the primitive-recursive listing, the f_m indices are computable uniformly in m. Thus all thresholds are uniformly Π^0_3. Moreover

$$H_{\mathrm{PR}}(p) = \infty \quad \Longleftrightarrow \quad \forall m\ T(p, f_m)$$

is still Π^0_3: the extra universal quantifier joins the first block. Oracle $0'''$ first decides this predicate. In the finite case it tests thresholds until the first failure and returns one less than that position. This proves the upper bound for the original measure and, by computing two values, for its comparator.

4 Prediction along the actual history

The lower bound will deliberately flip a good baseline's predictions. This changes the evader's future input. It is therefore insufficient to compare the new predictor with a good predictor's unmodified play. We need a bounded prediction rule whose correctness is independent of how the past predictions were chosen.

Fix a total computable time budget b. Given actual paired histories $x_{<n}$ and $y_{<n}$, examine program indices $i \le n$. A candidate is eligible when:

1. at each earlier round $r < n$, if its computation on $y_{<r}$ halts in fewer than $b(r)$ steps, its output equals x_r;

2. on $y_{<n}$ it halts in fewer than $b(n)$ steps with a binary output.

Choose the least eligible index and predict its output, or predict zero if none is eligible. Call this rule B_b. A timeout in an earlier test is ignored; it does not disqualify a candidate.

Lemma 4.1 (Robust baseline)**.** *If e is a total binary evader with $t_e(n) < b(n)$ eventually, then B_b predicts $e(y_{<n})$ correctly at all but finitely many rounds, regardless of the sequence of actual past predictions y.*

Proof. Let k be an index of e. Once n is large enough, $k \leq n$ and the current computation of e is within budget. Every earlier bounded computation of e agrees with the recorded x_r, since $x_r = e(y_{<r})$ in the actual play. Thus e is eligible from then on, and the least eligible index is at most k.

If a selected index $i < k$ gives a wrong output at round n, that bounded computation on the recorded input $y_{<n}$ permanently contradicts the earlier-round test at every later round. It can never be selected again. There are finitely many indices below k. After their possible first wrong selections, every selected output is correct. The proof never assumed that the actor actually used the baseline bit as its own prediction. □

To implement this rule in an ordinary predictor whose input is just $x_{<n}$, compute its own previous outputs on the strictly shorter prefixes $x_{<r}$, $r < n$. They reconstruct $y_{<n}$. Structural recursion on input length terminates. This may be computationally expensive, but it defines a total program, which is all the degree argument requires. It remains total if a decidable rule flips its current baseline output.

4.1 An infinite-level predictor for the original list

For the primitive-recursive hierarchy set

$$b_\infty(n) = f_{n+1}(n).$$

It is an ordinary total computable function, since the list has an effective total evaluator. It need not itself be primitive recursive. For every fixed m, $b_\infty(n) \geq f_m(n)$ eventually. The predictor which recursively reconstructs its own history and always uses B_{b_∞} therefore learns every evader in every E_{f_m}. Denote this fixed total predictor by q_∞; it satisfies

$$H_{\mathrm{PR}}(q_\infty) = \infty. \tag{4.1}$$

5 Coding Π^0_3 information into finite and infinite errors

Choose a fixed list position m_0 containing the function $n \mapsto n+1$. Then $f_m(n) \geq n+1$ for every $m \geq m_0$. For each $k \in \mathbb{N}$, let e_k ignore input bits and output 1 if input length is less than k, and 0 otherwise. It reads at most k cells, so its runtime is bounded by a constant c_k, depending on k. Consequently

$$e_k \in E_{f_m} \quad \text{for every } k \text{ and } m \geq m_0. \tag{5.1}$$

Its evasion stream is $1^k 0^\omega$, whatever predictions it receives. No runtime bound uniform in k is claimed.

Let $B \subseteq \mathbb{N}$ be any Π^0_3 set. Use a recursive normal form

$$z \in B \quad \Longleftrightarrow \quad \forall k\ \exists s\ \forall t\ R_0(z,k,s,t).$$

Uniformly produce computably enumerable sets $W_{z,k}$ such that

$$z \in B \quad \iff \quad \forall k \ (W_{z,k} \text{ is finite}). \tag{5.2}$$

Here is a direct construction. At stage t, take the least $s \leq t$ passing all tests $R_0(z, k, s, u)$ for $u \leq t$, or use $t + 1$ if none does. Enumerate a fresh number whenever this candidate changes. If there is a least true witness, all smaller candidates eventually fail and the least true witness stabilizes. If no true witness exists, the candidate eventually exceeds each fixed bound and changes infinitely often. This proves (5.2).

Proposition 5.1. *For any fixed ordinary total bound b eventually exceeding each t_{e_k}, the predicate $T(p, b)$ is Π^0_3-hard on total predictors. If $b = b_\infty$, the same reduction makes membership in B equivalent to infinite original measure, and equivalent to every fixed threshold $m \geq m_0$.*

Proof. Build p_z as the robust B_b rule using its own reconstructed prediction history, except for this flip: if $x_{<n}$ begins with a completed code 1^k0, flip the baseline bit precisely when the stage-n enumeration of $W_{z,k}$ adds a new element. The first zero determines at most one k. The code is permanent once completed. The program is total by structural recursion and is produced computably from z, with the fixed ordinary bound program hardcoded.

If $z \in B$, every selected set $W_{z,k}$ is finite. On any play there are therefore finitely many flips; a stream of all ones never completes a code and has none. Against any $e \in E_b$, the robust baseline is eventually correct even with these modified predictions. Thus $T(p_z, b)$ holds.

If $z \notin B$, choose k with $W_{z,k}$ infinite. Against e_k, the stream completes precisely the code 1^k0. The baseline is eventually correct by the budget assumption, but there are infinitely many subsequent flips. They give infinitely many genuine errors, so $T(p_z, b)$ fails.

For $b = b_\infty$, a positive input is learned at every original level by the same robust argument, giving infinite measure. A negative input fails against $e_k \in E_{f_{m_0}}$, so its measure is less than m_0. This proves all the further equivalences. □

The threshold upper bounds and Proposition 5.1 prove the first two parts of Theorem 2.1. Reading a fixed complete threshold from the value proves the lower bound $0''' \leq_T H_{\mathrm{PR}}$. For the strict comparator, combine (4.1) with the reduction:

$$H_{\mathrm{PR}}(p_z) < H_{\mathrm{PR}}(q_\infty) \quad \iff \quad z \notin B.$$

An oracle for this relation decides a complete Π^0_3 set by complementing its answer. Together with the upper bounds this proves the remaining assertions. It does not require a predictor of every exact finite measure value.

6 The canonical list of all total computable functions

We now change the list, keeping the game, runtime convention, and formula for H_g unchanged. Fix an acceptable numbering φ_e of partial functions $\mathbb{N} \rightharpoonup \mathbb{N}$. Scan indices in increasing order, retain an index when it is total and its function differs from every earlier retained function. Let a_m be the mth retained index and $g_m = \varphi_{a_m}$.

Lemma 6.1. *The sequence $m \mapsto a_m$ is computable in $0''$. Each entry is an ordinary total computable function, and every such function occurs exactly once. Oracle $0''$ uniformly supplies an ordinary program index for f_m.*

Proof. Totality is a Π^0_2 property decided by $0''$. For two promised total programs, search for an argument where their outputs differ. This ordinary search halts exactly when they differ, so equality is decidable with $0'$. Each finite comparison with previously retained entries is therefore available in $0''$. The scan keeps exactly the least index of each total function. There are infinitely many distinct total functions, so each requested position is reached. Given the finitely many ordinary indices $a_1, \ldots, a_m$, compile an ordinary program taking the finite maximum in (2.1). It is total. Only the choice of its index uses the oracle; its evaluation does not. □

Lemma 6.2. *For any listing of all total computable functions, every total predictor has a finite H_g value.*

Proof. Given total p, recursively define $z_n = 1 - p(z_{<n})$. Let e_p ignore its input bits and output z_n on inputs of length n. This is a total binary evader, and along the actual play p is wrong at every round. Its worst-case runtime $t_{e_p}(n)$ is total computable: run it on the finitely many length-n inputs and take the maximum halting time. The function $t_{e_p} + 1$ therefore appears as some g_j. For every n, $f_j(n) \geq t_{e_p}(n) + 1$, so $e_p \in E_{f_j}$. This threshold fails, and monotonicity makes every later threshold fail too. □

Proof of Theorem 2.2. To compute $H_g(p)$, use $0'' \leq_T 0'''$ to obtain each ordinary f_m index from Lemma 6.1, and use the uniform formula of Proposition 3.3 with oracle $0'''$ to decide $T(p, f_m)$. Search for the first failure. Lemma 6.2 guarantees termination and gives the answer $m - 1$. This proves $H_g \leq_T 0'''$.

Choose the fixed list position m_0 of $n \mapsto n + 1$. For every fixed $m \geq m_0$, all code evaders e_k belong to E_{f_m}, by the same constant-runtime argument as (5.1). Hardcode the ordinary program for this fixed f_m into Proposition 5.1. It reduces an arbitrary Π^0_3 set to $T(p, f_m)$. The upper formula remains Π^0_3 for this fixed program, so the threshold is complete. Reading one such threshold from H_g proves the lower degree bound. All these reductions output ordinary total predictors. □

The noncomputable enumeration therefore does not force an extra jump in the measure. Its oracle is used to select an ordinary bound program; the threshold problem for that program was already uniformly third-level. Confusing this composition with a third jump *relative to* $0''$ would give an unnecessarily large upper bound.

More generally, if a listing supplies ordinary total function indices computably in an oracle E, the same argument gives

$$H_g \leq_T E \oplus 0'''.$$

This is an upper bound, not a claim that H_g can recover E. The finite-value lemma still applies whenever the list covers all total computable functions.

7 What is classified, and what remains representation dependent

The two measures have equal oracle degree but different ranges. The original list admits the infinite-level predictor q_∞; every value for the all-total-function list is finite. Degree alone does not erase this structural distinction.

The original source conjecture attributes a possible oracle increase to the noneffective total-function list. The specified canonical listing above shows that this increase is not automatic:

the measure remains third-jump computable and third-jump hard. An arbitrary noneffective reordering is not covered by its upper bound. Nor does a lower bound for a finite-valued measure by itself give the same lower bound for its two-input strict comparator; the original-measure comparator proof used an infinite-valued calibration predictor unavailable in the new list.

The runtime convention is also substantive. The code evaders use a constant amount of work for each fixed k. A model imposing a mandatory full-input scan would need a different common bound before the hardness proof could be transferred. Totality promises and the ordinary program-index representation should likewise remain visible when quoting the result.

Finally these are computability results for specified adversarial prediction measures. They do not claim that third-jump calculations can be implemented as practical intelligence tests, or settle the ordinal, hyperreal and other taxonomies considered in [1].

Preparation. This AI-assisted hand-proof manuscript includes the adaptive-history lemma, quantifier calculations and lower-bound constructions. Standard arithmetical-hierarchy and jump facts are background inputs. No formal certification or established priority is claimed.

References

[1] S. Alexander and B. Hibbard, *Measuring Intelligence and Growth Rate: Variations on Hibbard's Intelligence Measure*, arXiv:2101.12047v1, 25 January 2021. Especially Definitions 2–7, Propositions 8–9 and the complexity discussion following Proposition 9.